# Individual Project MGT 8180

**Grade (15 points out of 100)**

## Choose one dataset to work on: Bags Tax or Food Waste at GSU.

**Below is the list of tasks required for each dataset, along with the associated grades.**

# Individual Project: Analysis of Bag Tax Data in the U.S. Stores.

**Objective:** Analyze the provided **Bag Tax Data** to extract meaningful insights using **Excel/Python /Python in Excel/ or any other analysis software tool that you prefer**. The project is based on developing descriptive statistics, trend analysis, and predictive modeling.

Include any assumptions made or subsets of the data that you decided to investigate.

## Project Instructions

### Task 1: Data Exploration ( 3 Points)

- Load and inspect the dataset in Excel.
- Identify if any steps were made to handle missing or inconsistent data.
- Provide a summary of the dataset (e.g., number of records, key variables, date range).

### Task 2: Descriptive Analysis (3 Points)

- Calculate key summary statistics for **Bag Count.**
- Identify the top 10 vendors contributing the most in terms of **Bag Count** and **Amount Collected**.
- Create **two visualizations** that you find useful based on the dataset explored.

### Task 3: Trend Analysis (3 Points)

- Create a **pivot table** to display any insightful information found (Sampling method is up to you; specify the selected range and the rationale for that selection).
- Show how revenue collection has changed over time. (You decide the scope (i.e., State level, store level, time frame..etc))

### Task 4: Predictive Analysis (3 Points)

- Perform r**egression analysis**.
- Use **time series forecasting**.
- Present results with appropriate graphs (trendline, regression plots, etc.).

### Task 5: Business Insights and Recommendations (3 Points)

- Summarize findings.
- Provide actionable recommendations based on data insights (e.g., tax policy adjustments and vendor trends).

**Deliverable:** Word file with a screenshot of the charts or analysis, along with your insights and an illustration of the results.

# Individual Project: Food Waste at GSU

**Objective:** Analyze the provided **Central Dining Hall** files to extract meaningful insights using **Excel/Python /Python in Excel/ or any other analysis software tool that you prefer**. The project is based on developing descriptive statistics, trend analysis, and predictive modeling.

Include any assumptions made or subsets of the data that you decided to investigate.

## Project Instructions

### Task 1: Data Exploration ( 3 Points)

- Load and inspect the dataset in Excel (There are various files, so you will need to check and select which files you will work with based on your interest and the questions that you are investigating).
- Identify if any steps were made to handle missing or inconsistent data.
- Provide a summary of the dataset (e.g., number of records, key variables, date range).

### Task 2: Descriptive Analysis (3 Points)

- Calculate key summary statistics**.**
- Identify the top 10 **wasted items** and the **amount wasted**.
- Create **two visualizations** that you find useful based on the dataset explored.

### Task 3: Trend Analysis (3 Points)

- Create a **pivot table** to display any insightful information found (Sampling method is up to you; specify the selected range and the rationale for that selection).
- Show how food waste has changed over time. (You decide the scope (i.e., Total food waste, or certain food item(s), time frame, etc.))

### Task 4: Predictive Analysis (3 Points)

- Perform r**egression analysis**.
- Use **time series forecasting**.
- Present results with appropriate graphs (trendline, regression plots, etc.).

### Task 5: Business Insights and Recommendations (3 Points)

- Summarize findings.
- Provide actionable recommendations based on data insights (e.g., What are your recommendations for GSU dining hall based on the analysis so that they reduce food waste?).

**Deliverable:** Word file with a screenshot of the charts or analysis, along with your insights and an illustration of the results.